



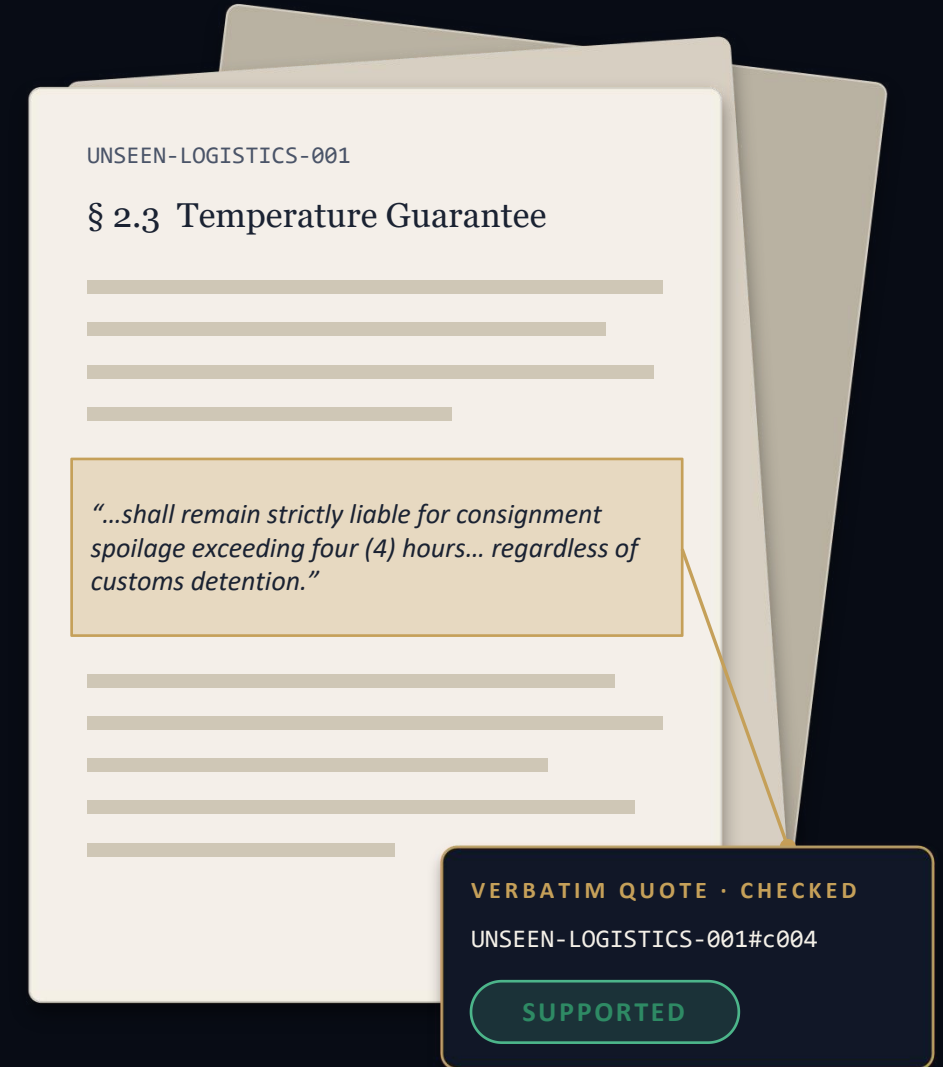
HNX Legal Intelligence
EVIDENCE-GROUNDED CONTRACT ANALYSIS

HACKNEX 2026 · HNX26EPS01 — AGENTIC LEGAL ASSISTANT

Legal AI must prove what it says

HNX Legal Intelligence — evidence-grounded analysis for complex commercial contracts

A fluent answer is not the same as a reliable legal answer.



A fluent answer is not a verifiable answer

01

Obligations are scattered

Duties, carve-outs and schedules sit in different sections.

02

Exceptions change the answer

“Subject to” and “notwithstanding” can reverse a general rule.

03

Fluency hides missing support

A generic model can answer confidently with no source behind it.

04

Silence matters

What the contract does not say is often the key finding.

FLUENT ANSWER · ILLUSTRATIVE

“The provider is not liable for delays caused by customs.”

NO SOURCE

EXCEPTION MISSED

VERIFIABLE ANSWER

Not liable for customs-clearance delays (**§2.2**) — except for Temperature Sensitive Goods: strictly liable for spoilage beyond four hours of the delivery window, regardless of customs detention (**§2.3 overrides §2.2**).

§2.2 QUOTED

§2.3 QUOTED

SUPPORTED

Clauses from UNSEEN-LOGISTICS-001; retrieval and the override link verified live (slide 5).

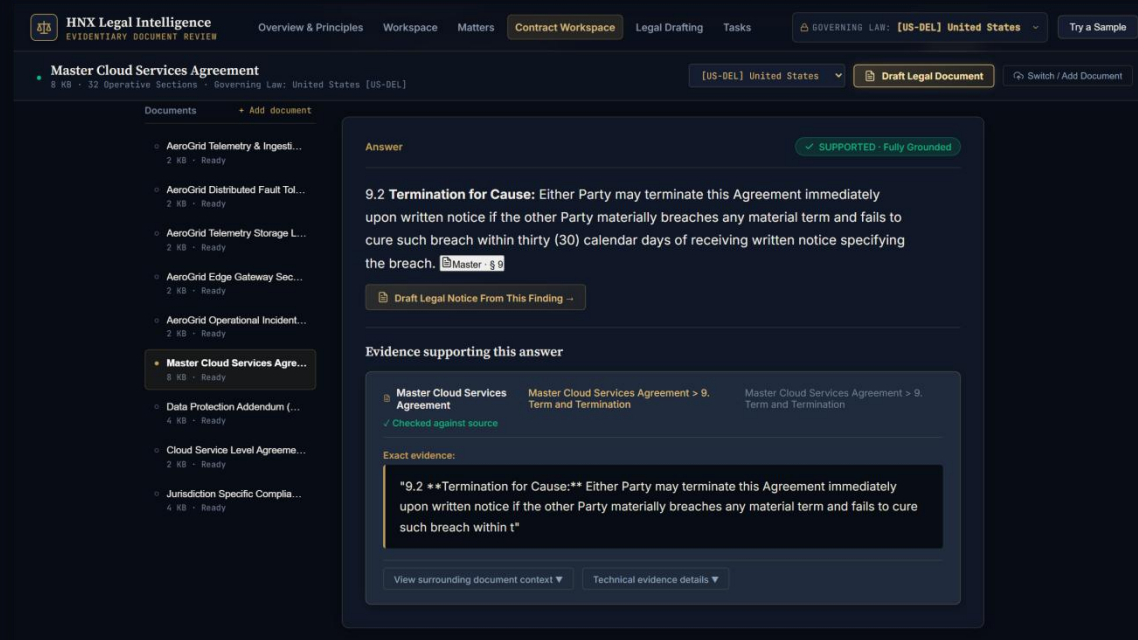
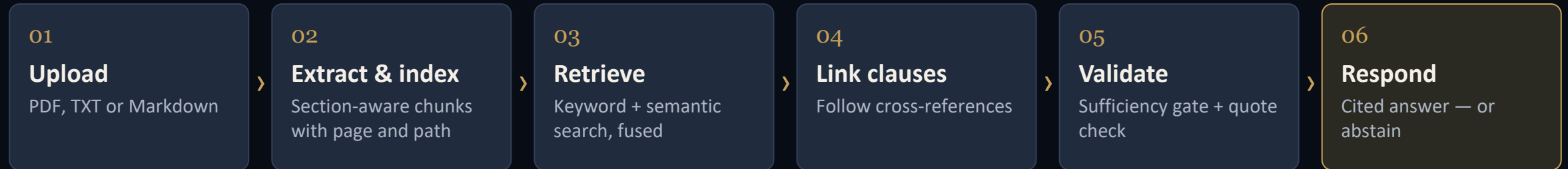


Three ways to read a contract

	Manual review	Generic LLM document chat	HNX Legal Intelligence
Evidence trail	Strong, but slow to assemble	Sources often missing or hard to check	Every cited quote is checked against the source text
Cross-clause links	Depends on reviewer attention	Left to the model; not inspectable	Follows “subject to”, “notwithstanding”, “defined in”
Missing evidence	Reviewer notes the gap	May fill the gap with plausible text	Abstains — no answer, no citations
Drafting	Written from scratch	Facts and allegations can blur	Clauses quoted, allegations labelled, gaps left as placeholders
Lawyer review	Is the review	Still required	Still required — drafts are labelled for review

An answer without supporting evidence is not enough for legal work.

One workflow, from upload to evidence



- 1 **Status in words, not just colour**
“SUPPORTED · Fully Grounded” or “INSUFFICIENT · Evidence Not Established”.
- 2 **Exact evidence, with its location**
The quoted span, the document and the section path.
- 3 **From finding to draft**
One click opens a notice draft labelled for lawyer review.

Live product — Contract Workspace, captured 9 Oct 2026



A contract it has never seen

UNSEEN-LOGISTICS-001 · synthetic cold-chain pharmaceutical logistics agreement · uploaded during the demo, questions scoped to this document

1 · CROSS-CLAUSE QUESTION

SUPPORTED

“Is the service provider liable for spoilage of temperature sensitive goods when customs detention delays delivery?”

§2.2 “Subject to Section 2.3, Service Provider shall not be liable for delivery delays attributable to port customs clearance operations.”

§2.3 “Notwithstanding Section 2.2, ... shall remain strictly liable for consignment spoilage exceeding four (4) hours beyond the scheduled delivery window regardless of customs detention.”

DETECTED: §2.3 OVERRIDES §2.2

Customs delays are excused in general — but not spoilage of temperature-sensitive goods beyond four hours.

2 · OUT-OF-SCOPE QUESTION

“What are the aircraft hangar maintenance guidelines and hazardous chemical disposal protocols?”

INSUFFICIENT

0 citations. The system names the missing subject matter instead of guessing.

ALSO VERIFIED

Standard delivery window → **“forty-eight (48) hours”**
§2.1, quoted verbatim

Checked against the running system on 9 Oct 2026 (/api/ask, scoped by doc_id). Gemini was unavailable at capture time, so answers came from the extractive path: verbatim source sentences.

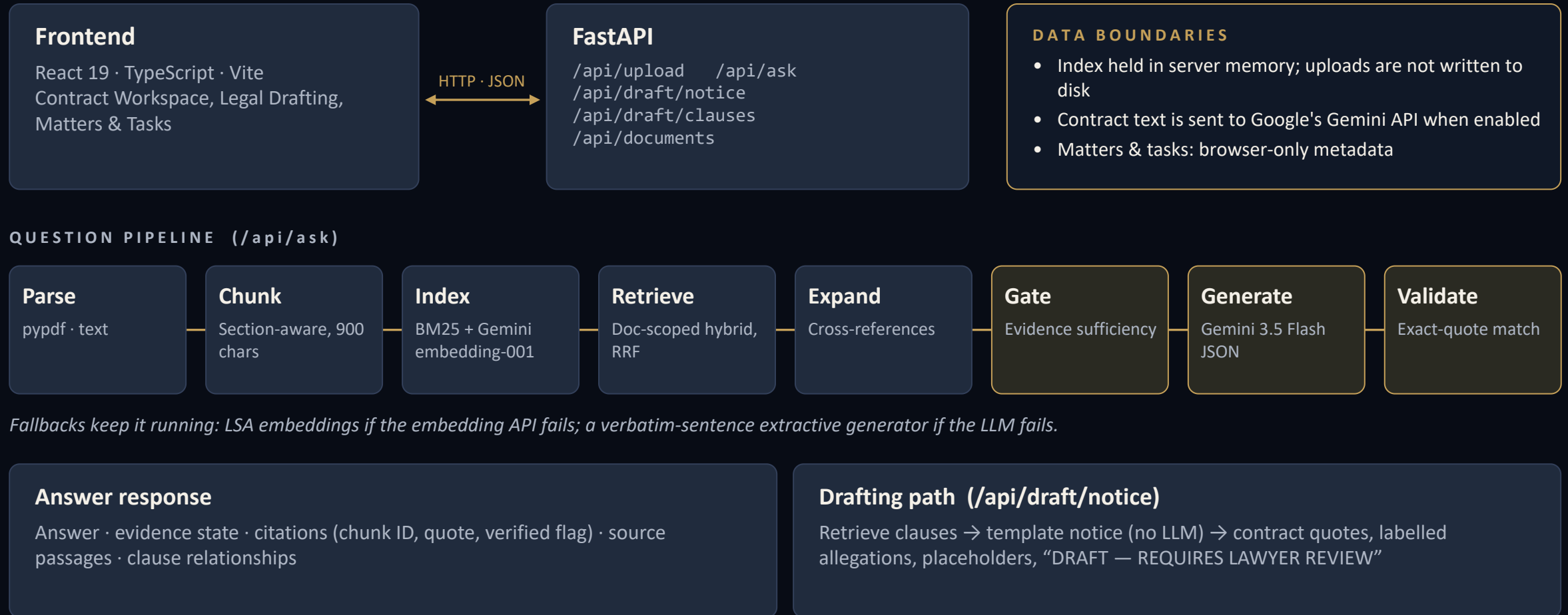


Evidence integrity, layer by layer

LAYER	WHAT IT DOES	WHY IT MATTERS
1 Document-scoped retrieval	Searches only the selected contract	Answers aren't borrowed from other files
2 Keyword search (BM25)	Matches exact terms and section numbers	Legal wording is precise
3 Semantic search	Gemini embeddings find passages by meaning	Catches paraphrased obligations
4 Reciprocal Rank Fusion	Merges both rankings by rank ($k = 60$)	No fragile score calibration
5 Clause-dependency expansion	Adds up to 2 clauses named by “subject to”, “notwithstanding”, “defined in”	Exceptions travel with the rule
6 Sufficiency gate + quote check	Abstains when core terms are absent; cited quotes must match the source exactly	No evidence → no answer, no citation



How a question becomes a cited answer



Evaluated, not asserted

Real-LLM baseline · plain hybrid RAG

Gemini 3.5 Flash + gemini-embedding-001 · 16 questions (8 answerable, 4 unanswerable, 4 adversarial) · internal technical corpus

100%

Recall@4
right document retrieved

15/16

Abstention decisions
correct

0

Citations to chunks
outside the retrieved set

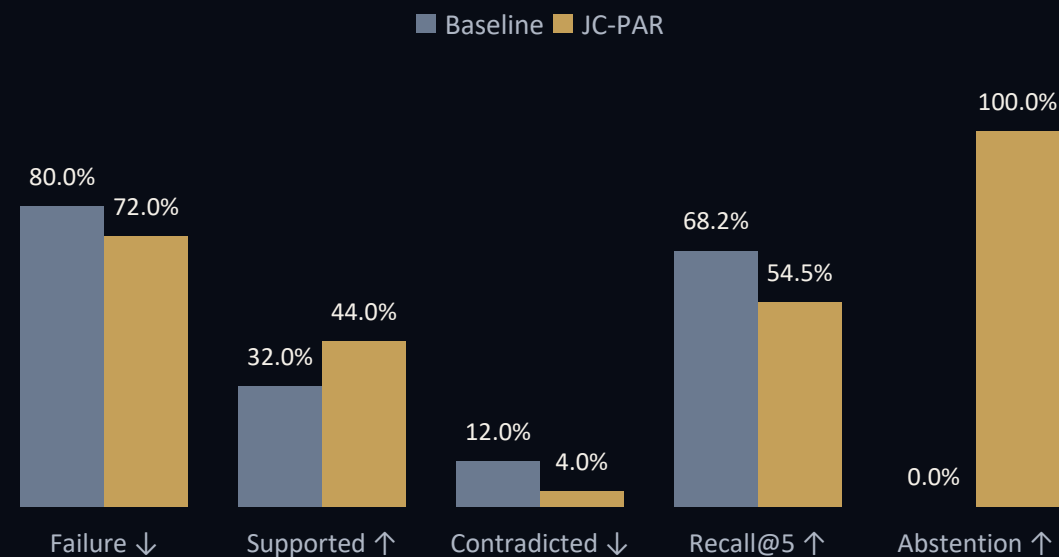
25/28

Cited quotes
exact match to source

34 / 34 automated tests pass (9 Oct) — including 10 unseen-contract scenarios: verbatim spans, §2.3↔§2.2 links, abstention, document isolation, injected-instruction text.

Adversarial legal stress test · 25 cases

Baseline RAG vs JC-PAR (jurisdiction routing + clause expansion + sufficiency gate)



↓ lower is better · ↑ higher is better. Supported = strictly supported answers; abstention = correct on 3 unanswerable cases (0/3 → 3/3). Extractive generator, 8 Oct code snapshot. Recall@5 fell — an open problem we report, not hide.

Not yet measured: claim-level entailment · lawyer-rated usefulness · real-LLM run on the legal stress set



Useful today, honest about its limits

INTENDED VALUE

Design goals — not yet measured with lawyers

- Jump to the clause that answers the question
- Trace each statement to a quoted source
- See exceptions next to the rule they modify
- Know when the contract is silent
- Start notices from a lawyer-review draft

LIMITATIONS

What we tell users up front

- A lawyer must still review every output
- An exact quote isn't proof the claim is right
- Conflict flags rely on the model's classification
- Clause expansion lowered Recall@5 (68% → 55%)
- PDF, TXT and Markdown only — no DOCX or OCR yet

ROADMAP

BUILT

Hybrid RRF retrieval · clause expansion · sufficiency gate · quote validation · template drafting · matters & tasks prototype

NEXT

- Claim-level entailment checks
- Rule-based conflict flag in every answer
- Fix the expansion recall penalty
- DOCX and OCR ingestion
- Evaluation with practising lawyers

Built to help lawyers verify faster — not to replace their judgment.



CLOSING

Don't just generate an answer. Show why it deserves to be trusted.



HNX treats every legal answer as a claim that must carry its evidence — or say plainly that the evidence isn't there.

HNX Legal Intelligence

Team No Sleep Till Deploy

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Ask us anything about the unseen contract — then inspect the evidence trail.

Evaluation runs and their caveats

Run	What ran	Data	Headline results	Caveats
Real-LLM baseline · 8 Oct	Plain hybrid RAG, top-4; Gemini 3.5 Flash; gemini-embedding-001 (3072-d)	16 questions; 9 internal technical documents, 52 chunks	Recall@4 100% · groundedness 100%* · usefulness 85%* · abstention 15/16 · out-of-set citations 0	*lexical heuristics, not entailment; document-level recall; 3/28 quotes not exact; false-premise QA-015 wrongly abstained
Adversarial stress test · 8 Oct	Baseline vs JC-PAR (JARF + CPDE + ESV); extractive generator	25 legal cases (3 unanswerable, 5 jurisdiction)	Failure 80% → 72% · abstention 0/3 → 3/3 · contradictions 3 → 1 · Recall@5 68.2% → 54.5%	No LLM; earlier code snapshot; jurisdiction accuracy 2/5 for every variant
Automated tests · 9 Oct	pytest, 9 test files	34 tests incl. 10 unseen-contract scenarios	34 passed in 13.8 s	Most unseen-contract tests use the extractive path; the conflict test accepts several states
Live demo check · 9 Oct	/api/ask on UNSEEN-LOGISTICS-001, scoped by doc_id	4 questions	3 answered with verbatim quotes (status SUPPORTED) · 1 correct abstention, 0 citations	Gemini unavailable → extractive path; payment-terms answer missed the 30-day term

Sources: data/results/baseline_gemini_hybrid_rrf.json · data/results/ablation/*.json · pytest run of 9 Oct 2026 · presentation/assets/demo_verification.json



Contract Workspace on UNSEEN-LOGISTICS-001: the cross-clause question returns SUPPORTED with §2.3, §2 and §5 as evidence sources.

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Legal Drafting: notice generated from retrieved clauses, with the DRAFT — REQUIRES LAWYER REVIEW banner and a separate contract-evidence panel.